Golang RSS Reader service and Rails RSS Reader application

If you've ever used Google Reader, Feedly, Digg or any other RSS reader, then you must be familiar with what RSS readers generally do.
For the sake of assignment purpose, please create an RSS reader by taking the following steps:

1. Create a Golang RSS Reader package, which can parse asynchronous multiple RSS feeds.

2. Create a Golang Rss Reader service that uses the RSS package

3. Create a Rails application that contains the following features:

   3.1. Store, manage and display the RSS URLs.

   3.2. Store, manage and display the RSS posts.

   3.3. Requests the Golang RSS Reader to parse a set of URLs.

RSS Reader service

RSS Reader package

Use the latest stable version of Go.
The package must have only exportable type `RssItem` and method `Parse` that:

1. Takes an array of URLs.
2. Parses asynchronously their feed.
3. Returns an array of `RssItem` generated from all provided RSS posts.

```
type RssItem struct{
  Title        string
  Source       string
  SourceURL    string
  Link         string
  PublishDate  time.Time
  Description  string
}
```

Cover all changes with tests.

Add a README file with an explanation how the package should be installed and used.

RSS Reader service

Use the latest stable version of Go in order to implement service that handles JSON API requests that contain URLs.
Using the already created RSS Reader package, it will parse the URLs and return as JSON in the response the following structure

```
{
    "items": [
        {
            "title":        "<ProperValue>",
            "source":       "<ProperValue>",
            "source_url":   "<ProperValue>",
            "link":         "<ProperValue>",
            "publish_date": "<ProperValue>",
            "description":  "<ProperValue>"
        }
        ....
    ]
}
```

Add a README file with an explanation of set up and usage of the application.

Rails RSS Reader application

1. Use the latest stable `Rails` version.

2. View engine and Frontend Framework.

   2.1. Use `Slim` view engine.

   2.2. Bootstrap.

3. Create CRUD endpoint and UI for RSS URLs.

4. Create a posts preview page.

   4.1. Fetch and display latest posts.

   4.2. Sort by date with the most recent first.

   4.3. Be able to filter posts by RSS feed, Date of creation and Title.

   4.4. Display the titles only, clicking on a title will open original article page in new tab/window.

5. Background job that uses the *Golang RSS Reader service* to get and parse the posts.

6. Cover all changes with Rspec tests.

7. Add integration tests via *Capybara*.

Bonus points

We will give bonus points for the following:

- if the communication between Golang RSS Reader service and Rails RSS Reader application is via RabbitMQ or another messaging broker.
- if RSS Reader service implements some authorization for the requests, using JWT for example.
- if Rails RSS Reader application uses React.
- if both Ruby application and Go service are dockerized.

Additional Comments

1. Initial commit with all changes not directly related to the task – .gitignore file, etc.
2. All subsequent commits should be logically organized reflecting the steps you've taken developing the application – neither one large commit with all changes nor a multitude of smaller commits for every little tiny change.
3. Document your code where needed and add a short README.